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ROAD ACCIDENT RISK PREDICTION USING SUPERVISED MACHINE LEARNING TECHNIQUES

TY B.Tech. MDM Project Implementation Report (SEM V)

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SCHOOL OF COMPUTER ENGINEERING

MIT ACADEMY OF ENGINEERING, ALANDI (D), PUNE-412105

MAHARASHTRA (INDIA)

DECEMBER 2025

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Submitted in partial fulfilment of the
requirements for the award of the degree

of

Bachelor of Technology

in

ELECTRONICS & TELECOMMUNICATION ENGINEERING

GUIDED BY

Dr. Sunita Barve

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CERTIFICATE

It is hereby certified that the work which is being presented in the SY B.Tech. Project Implementation Report entitled **"*Road Accident Risk Prediction Using Supervised Machine Learning Technique*",** in partial fulfillment of the requirements for the award of the **Bachelor of Technology in Electronics & Telecommunication Engineering** and submitted to the **Department of Computer Engineering of MIT Academy of Engineering, Alandi(D), Pune, Affiliated to Savitribai Phule Pune University (SPPU), Pune** is an authentic record of work carried out during an Academic Year 2025-2026, under the supervision of **Dr. Sunita Barve, Department of Computer Engineering.**

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We would be failing in our duty if we did not thank all the other staff and faculty members for their experienced advice and evergreen cooperation

1. Aryan Kumar
2. Amir Furquani
3. Sakshi Patil

ABSTRACT

According to World Health Organisation (WHO) data, road accidents cause about 1.19 million deaths per year, costing around 3% of most countries' GDP. Worldwide road accident statistics show one death every 26 seconds globally. This paper targets on two main objectives: (1) to identify the major causes of road accidents by performing exploratory data analysis (EDA), and (2) to use supervised Machine Learning (ML) algorithms with hyperparameter tuning to predict the risk of a road accident. The model is trained and tested with 517,755 entries and 12 features, such as road curvature, speed limit, weather, and time of day. The proposed model employs supervised Machine Learning algorithms, such as Linear Regression, Polynomial Regression, Random Forest, XGBoost, and CatBoost. The experimental results demonstrate that the CatBoost model gives the lowest RMSE score (0.047), followed by XGBoost (0.051), showing its superior performance in predicting the probability of a road accident.

LIST OF FIGURES

Figure No.	Fig. Name	Page No.
Figure 1	Graphical Abstract of Proposed System	14
Figure 2	The first 5 rows of the dataset	15
Figure 3	Standard Scaler Applied on Curvature	16
Figure 4	Standard Scaler on Speed Limit	17
Figure 5	Correlation- Heatmap	17
Figure 6	Count of Different Categorical Columns	18
Figure 7	Outliers for Different Categorical Columns	19
Figure 8	Outliers for Different Numerical Columns	19
Figure 9	Reported Accident Vs Speed Limit	20
Figure 10	Curvature Vs Accident Risk	20
Figure 11	Curvature Vs Speed Limit	21
Figure 12	Actual vs Predicted Accident Risk for Linear Regression showing the best-fit relationship between true values and model outputs.	22

Figure 13	Actual vs Predicted Accident Risk for Polynomial Regression showing the best-fit relationship between true values and model outputs.	23
Figure 14	Actual vs Predicted Accident Risk for the KNN Regression Model	24
Figure 15	Feature importance ranking generated by the Random Forest model, showing the most influential variables for accident risk prediction	25
Figure 16	Learning curve showing the change in training and validation error across boosting iterations for the Gradient Boosting model.	26
Figure 17	Residual distribution for the CatBoost model showing prediction error concentration around zero, indicating strong model performance.	27
Figure 18	SHAP summary plot illustrating the global impact of each feature on LightGBM accident risk predictions.	28
Figure 19	Comparison of Different Model Parameters	32
Figure 20	Evaluation matrix of all model comparison tables, where a- R^2 of all models, b- MAE of all models, c- RMSE of all models and d- Accuracy of all models	33
Figure 21	Relation of Different Matrices	34
Figure 22	Landing Page After Deployment	34
Figure 23	Model Making Predictions	35

LIST OF TABLES

Table. No.	Table. Name	Page No.
Table 1	Evaluation Parameters of Different Supervised ML Models	31

CONTENTS

Acknowledgements			4
Abstract			5
List of Figures			6
List of Tables			8
1.	Introduction		10
	1.1	Motivation for the project	11
	1.2	Problem Statement	11
	1.3	Objectives and Scope	11
2.	Literature Survey		12
3.	Proposed System		14
	3.1	Dataset Details	14
	3.2	Feature Engineering	15
	3.3	Data Pre-Processing	16
	3.4	Exploratory Data Analysis	17
	3.5	Model Training and Testing	21
	3.6	System Interface Design and Deployment Architecture	29
	3.7	Results and Discussion	31
4.	Conclusion and Future Scope		36
5.	References		37

1. INTRODUCTION

Road accidents are a major global problem. They cause 1.19 million deaths each year and lead to economic losses of about 3% of GDP in many countries, according to the World Health Organisation (WHO). The growing amount of traffic, rapid urban growth, and human mistakes have increased the need for better accident-prevention systems. Traditional road safety analysis often relies on past trends and manual investigations. These methods often overlook the complex interactions between road conditions, the environment, and driver behaviour.

Recent studies indicate that Machine Learning (ML) can predict accident patterns and assess severity more effectively. Previous research has focused on ML-based severity prediction. Models like Random Forest, Logistic Regression, and SVM have performed well on regional accident datasets [2], [4], [5], [8], [11]. Other studies highlight the effectiveness of regression-based and hybrid models in predicting accident frequency, including Linear Regression, ANN, and ensemble methods [3], [7], [10], [14]. Large-scale studies from the UK, USA, and India consistently show that factors like weather, lighting, road type, and driver demographics significantly influence accident occurrence [5], [12], [16].

Despite extensive research on predicting severity, there is a noticeable gap in predicting accident risk, which is the likelihood of accidents. This gap exists even with high-volume datasets that involve thorough preprocessing, feature engineering, and well-tuned supervised ML models. This project aims to address that gap by creating a robust Road Accident Risk Prediction system using a dataset of 517,755 records. It will involve data preprocessing, feature engineering, train-test modelling, and deployment for real-time use.

1.1 Motivations

Road accidents remain a critical global issue, claiming over 1.19 million lives annually and causing substantial economic losses, particularly in developing nations. Traditional accident analysis methods rely heavily on historical trends and manual investigation, which often fail to capture the complex interactions between environmental conditions, road infrastructure, and driver behaviour. With the rapid increase in traffic volume and urbanisation, there is a growing need for advanced predictive systems that can estimate accident risk before incidents occur. Recent research shows that machine learning offers strong potential for improving predictive accuracy, yet most existing studies focus only on accident severity rather than the broader and more impactful task of predicting accident likelihood. This gap highlights the importance of developing a data-driven system that can proactively identify high-risk scenarios and support timely road safety interventions.

1.2 Problem Statement

- To develop a predictive model that accurately estimates the likelihood of road accidents using large-scale traffic, environmental, and road-condition data.
- To overcome the limitations of traditional analysis by incorporating machine learning techniques capable of modelling non-linear relationships among multiple accident-related factors.
- To address the research gap where previous models predict only severity, not risk, despite the availability of extensive datasets and advanced ML methods.
- To enable real-time accident-risk prediction by deploying the trained model in a scalable and practical system for end-user applications.

1.3 Objectives and Scope

The primary objective of this project is to design and develop a robust Road Accident Risk Prediction system using supervised machine learning techniques on a dataset of 517,755 records. The work involves extensive data preprocessing, feature engineering, model training, evaluation, and selection based on performance metrics such as RMSE [10, 14], MAE, R^2 [14], and accuracy. The scope of the system includes implementing multiple ML algorithms, comparing their effectiveness, and deploying the best-performing model for real-time accessibility through a user-friendly interface. Additionally, the project aims to create interactive visual analytics, enabling traffic authorities, policymakers, and researchers to understand risk patterns, identify influential factors, and make informed decisions. The system's scope is limited to predicting accident likelihood, not real-time accident detection or driver behaviour monitoring, ensuring a focused and achievable solution within the available dataset and computational constraints.

2 LITERATURE SURVEY

(Ahmed, Shakil, et al.) [2] Worked on the classification dataset, which contains 66403 accident records between 2016-2020 in the South Island, New Zealand, which also focused on features like Alcohol Suspected, Drugs Involved, and Driver's Age. This shows that Random Forest performs better than Linear Regression, KNN, and Naive Bayes for both binary and multiclass classification, with the highest accuracy of 86.66% and 67.67% respectively.

(S. Alagarsamy, rt al.) [3] worked on the UK Department for Transport (2018) record, which contains 220,000 accident records. Their Study shows that Random Forest combined with Linear regression gives the best accuracy of 97.86% compared to Decision tree and SVM.

(Krishna, U. Vivek, et al.) [4] Also worked on the classification dataset, using Supervised ML Algorithms such as Random Forest (88.89%), which gives the highest accuracy, Decision Tree with Hyperparameter tuning (85.74%), and Logistic Regression (86.23%).

(Vijithasena, Rasika, et al.) [5] worked on a classification dataset collected from 49 states of the USA between February 2016 to December 2019, which contains around 3 million accident records. Focused on the severity range (1-4), Random Forest gives the Highest accuracy of (97.2%)

(Ahsan, Annesha, et al.) [6] worked on a classification dataset with 1000 records. Got the highest accuracy with Decision Tree (95%) and Random Forest (93%).

(Mahendra, G. et al.) [7] worked on a regression dataset with 300 records collected from the Department of Transportation in 2016 across different locations of India. This Article focuses on the intensity of collision with attributes like Head-On Collision, Single-Lane, Fog Weather and Pedestrian Crossing. For this dataset Decision tree worked better than the Random Forest.

(Mustafa, Dheya et al.) [8] worked on the classification dataset collected from the Jordanian Traffic Department, which contains 73k real accident records. Model trained and tested on algorithms like SVM, KNN, NB, LR and DT, out of which LR gives the best accuracy of 97.80% followed by SVM 92.75%.

(Kaliraja, C, et al.) [9] took the dataset from Kaggle with general information of traffic count between years 2001 to 2014 with 1048576 rows and 32 columns. The article uses 2 models- Linear Regression (92%) and Random Forest (89%).

(Vinoth, B. et al.) [10] Used Linear, non-linear model approaches and Neural networks, using data of road accidents between 1994-2019. ANN worked best with the least RMSE of 17956.30.

(Dia, Yoro, et al.) [11] worked on the classification dataset of 2022. Used several supervised ML models such as random forest, KNN, Logistic regression, SVM and Gaussian NB, out of which Random Forest gives the highest accuracy (85.6%) followed by SVM. Focuses on the Severity of road accidents, with parameters such as vehicle and road environment characteristics.

(Paul, Arindam Kumar, et al.) [12] Using characteristics like Road Type, Light Condition, Weather Condition, Driver Age, Vehicle Type, and Carriageway Hazards, the Authors examined one million road accident records from the United Kingdom. They determined that the most important factors influencing accident severity were light condition, driver age, and carriageway hazards using Neighbourhood Component Analysis (NCA), K-Nearest Neighbours (KNN), and Partial Dependence Plots (PDP). An SVM model was used to validate the results, demonstrating high accuracy and confirming the efficacy of these feature-selection techniques.

(Manisha, V., et al.) [13] Analysed a Kaggle traffic dataset containing Date, Time, Zone, Weather, Temperature, and Traffic Volume from urban and rural areas. They compared SVM, Decision Tree, and Random Forest for accident prediction. Random Forest achieved the best performance, with 92% accuracy, 89.81% precision, and 83% recall, proving its superiority over SVM and Decision Tree models.

(Reddy, Jambula Snehith, et al.) [14] The Ministry of Road Transport & Highways and the Motor Vehicle Accident Fund Trust provided the national and state highway data for India (1990–2022). State-by-state accident statistics, vehicle and road types, weather, and urban/rural classification were all included in the dataset. To predict the frequency and severity of accidents, they tested several models, including Linear Regression, Ridge Regression, Decision Tree, KNN, and Multilayer Perceptron (MLP). With an R2 of 0.93278 and an RMSE of 0.3879, MLP performed the best, making it the most successful model for predicting traffic accidents.

(Augustine, Teres, et al.) [15] analysed government-recorded accident data (2018–2020) from an Indian district using features such as Time of Accident, Driver's Age, Vehicle Age, Road Type, and Weather Condition. They implemented Logistic Regression, Random Forest, Decision Tree, KNN, XGBoost, and SVM to predict accident occurrences. Random Forest achieved the best performance with 80.78% accuracy, effectively identifying accident-prone areas.

(Geyik, Buket, et al.) [16] Road Type, Light Condition, Weather Condition, Road Surface Condition, Urban/Rural Area, Junction Detail, Carriageway Hazards, and Pedestrian Crossing Type are among the 32 features that are used in their analysis of the STATS19 dataset, which included 280,443 UK traffic accident records from 2010 to 2012. Using Decision Tree, Random Forest, Naïve Bayes, and Multilayer Perceptron (MLP), accidents were categorised as Fatal, Serious, and Slight. The most successful method for predicting severity was MLP, which outperformed Random Forest (85.19%), Naïve Bayes (83.40%), and Decision Tree (80.74%) with the highest accuracy (86.67%).

3 PROPOSED SYSTEM

Figure 1 illustrates the overall architecture of the proposed Road Accident Risk Prediction System, which integrates data preprocessing, feature engineering, and supervised machine learning to accurately estimate accident risk levels. The system begins with the acquisition of a large-scale accident dataset, followed by preprocessing steps such as Exploratory Data Analysis (EDA), Standard Scaling, and One-Hot Encoding to ensure uniformity and improve model learning. Since the dataset contains no missing values, the preprocessing primarily focuses on transforming numerical features, encoding categorical variables, and preparing clean input for the model. Feature engineering is then applied to derive meaningful patterns from attributes such as road curvature, weather, speed limit, and time of day. The refined dataset is subsequently used to train and test multiple machine learning algorithms for performance evaluation and accident risk prediction. Finally, the best-performing model is deployed as an interactive and scalable system for real-time accident risk assessment and decision support.

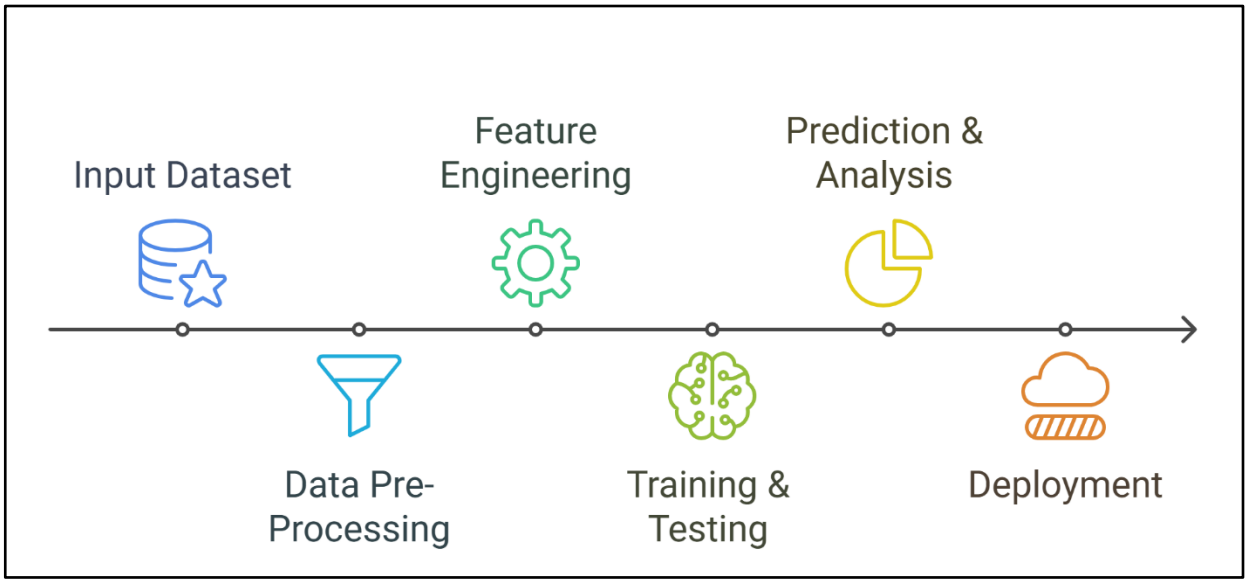


Figure 1: Graphical Abstract of Proposed System

3.1 Dataset Details-

The dataset used in this study was obtained directly from the Kaggle competition titled "Road Accident Risk Prediction," which permits the use of its data strictly for academic and research purposes. The dataset consists of 517,755 samples and 12 input features, such as Road Type, Speed Limit, Reported Accidents, etc., along with Accident Risk as the target variable.

These attributes collectively represent road infrastructure, environmental conditions, temporal factors, and historical accident information, making the dataset suitable for risk-based predictive modelling. The first 5 rows of the dataset are shown in Figure 2

	id	road_type	num_lanes	curvature	speed_limit	lighting	weather	road_signs_present	public_road	time_of_day	holiday	school_season	num_reported_accidents	accident_risk
0	0	urban	2	0.06	35	daylight	rainy	False	True	afternoon	False	True	1.0	0.13
1	1	urban	4	0.99	35	daylight	clear	True	False	evening	True	True	0.0	0.35
2	2	rural	4	0.63	70	dim	clear	False	True	morning	True	False	2.0	0.30
3	3	highway	4	0.07	35	dim	rainy	True	True	morning	False	False	1.0	0.21
4	4	rural	1	0.58	60	daylight	foggy	False	False	evening	True	False	1.0	0.56

FigurFigure 2: The first 5 rows of the dataset

3.2 Feature Engineering-

Feature engineering was performed to enhance the predictive power of the machine learning model by generating new variables that capture road geometry, traffic intensity, environmental conditions, and temporal risk patterns. Interaction features such as *curvature_speed*, *curvature_lanes*, *speed_lanes*, and *accidents_curvature* were created by multiplying key road attributes to represent their combined effect on accident occurrence. The formula used is:

$$\text{curvature_speed}=\text{curvature}\times\text{speed_limit} \text{ ---- } 1$$

$$\text{curvature_lanes}=\text{curvature}\times\text{num_lanes} \text{ ------} 2$$

$$\text{speed_lanes}=\text{speed_limit}\times\text{num_lanes} \text{ -----} 3$$

$$\text{accidents_curvature}=\text{num_reported_accidents}\times\text{curvature} \text{ ---} 4$$

Polynomial transformations (*curvature_squared* and *curvature_cubed*) were introduced to model nonlinear curvature effects.

$$\text{curvature_squared}=(\text{curvature})^2 \text{ -----} 5$$

$$\text{curvature_cubed}=(\text{curvature})^3 \text{ ----- } 6$$

Several binary risk indicators, including *high_curvature*, *high_speed*, *dangerous_combo*, and *poor_visibility,* were designed to highlight hazardous road and environmental situations.

$$high_curvature = \begin{cases} 1, if\ Curvature > 0.6 \\ 0, otherwise \end{cases} \text{-------} 7$$

$$high_speed = \begin{cases} 1, if\ speed_limit > 60 \\ 0, otherwise \end{cases} \text{------} 8$$

$$\text{dangerous_combo}=\text{high_curvature}\times\text{high_speed} \text{ -----} 9$$

Time-dependent features such as *rush_hour* and *peak_risk_time* capture increased risk during peak movement hours and holidays. A domain-driven score (*road_risk_score*) was assigned based on road type, while *accident_rate_per_lane* incorporated historical accident frequency normalised by lane count.

$$\text{accident_rate_per_lane} = \frac{\text{num_reported_accidents}}{(\text{num_lanes} + 1)} \quad \text{------ } 11$$

These engineered features collectively enrich the dataset with meaningful patterns, improving the model's ability to learn complex relationships and accurately predict accident risk.

3.3 Data Pre-Processing

Standard Scaling was applied to the numeric features of the dataset, such as Speed Limit, Road Curvature, and Number of Lanes, to normalise the range of values before model training. The Standard Scaler transforms each numeric feature using the formula:

$$z = \frac{x-\mu}{\sigma} \text{ ----- } 12$$

where $\mu$is the mean and $\sigma$is the standard deviation. This ensures that all numeric variables follow a zero-mean and unit-variance distribution, preventing features with large numerical ranges from dominating the model. Figure 3 and Figure 4 show the effect of Standard Scaling on selected numerical features, illustrating how the values become standardised and uniformly distributed after preprocessing.

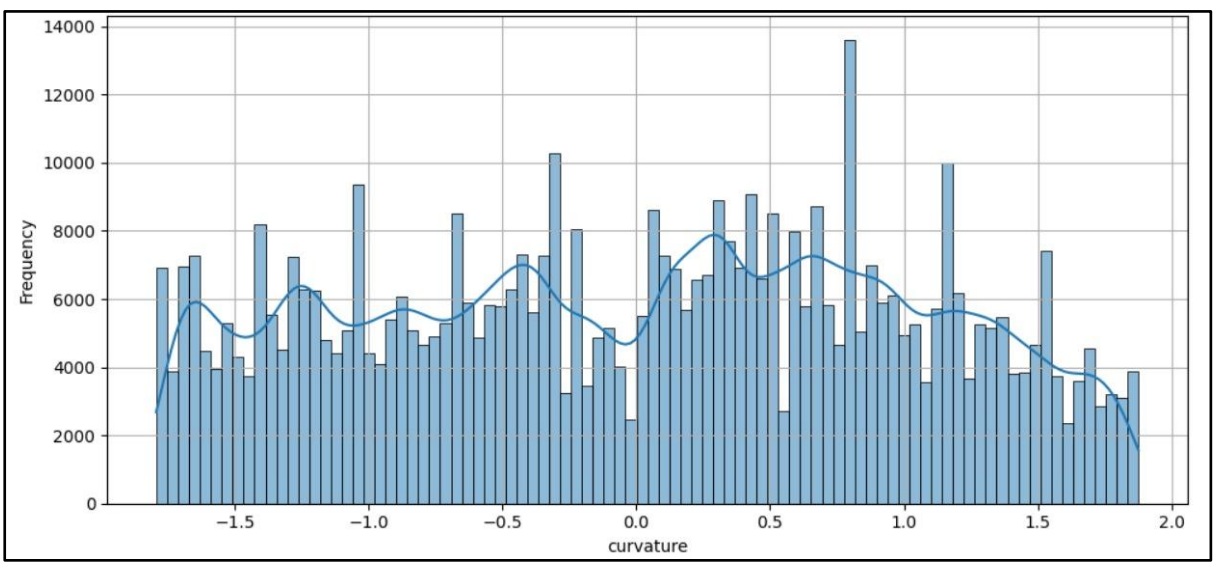


Figure 3- Standard Scaler Applied on Curvature

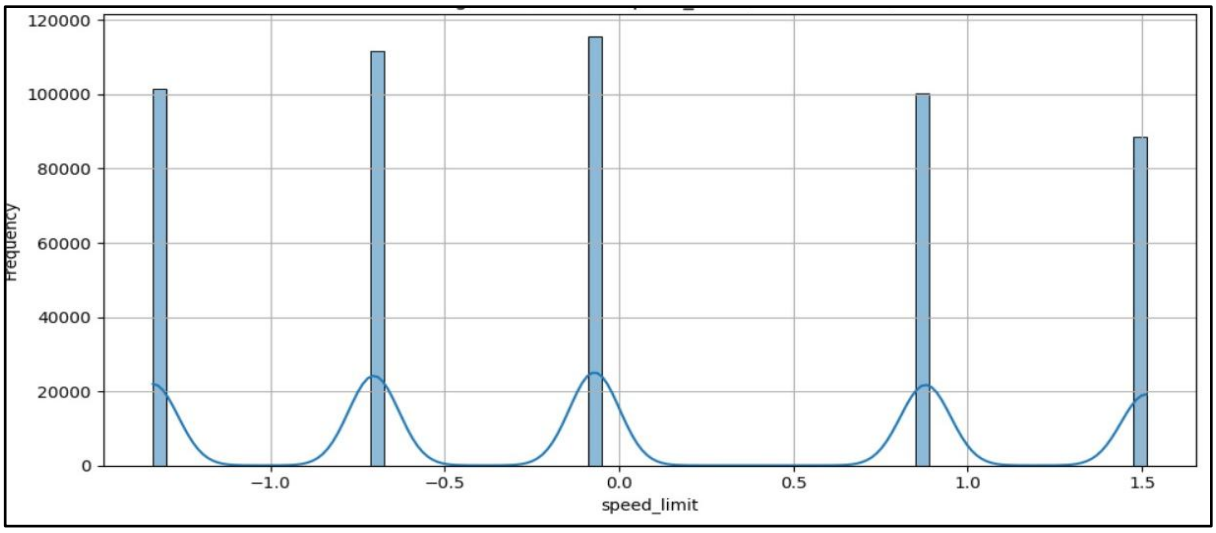


Figure 4- Standard Scaler on Speed Limit

One-Hot Encoding was applied to categorical features such as Road Type, Weather Condition, and Lighting Condition to convert non-numeric labels into machine-readable binary vectors. In this process, each category is transformed into a separate column with values 0 or 1, ensuring that the model does not assume any ordinal relationship between categories. This encoding allows algorithms to correctly interpret categorical variables without introducing unintended bias.

3.4 EDA

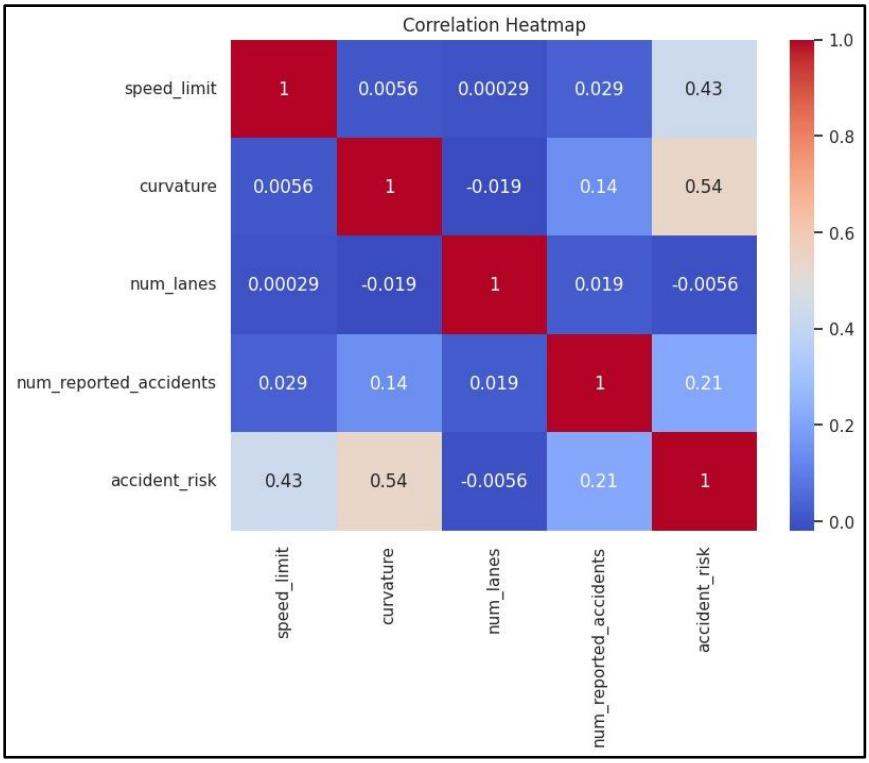
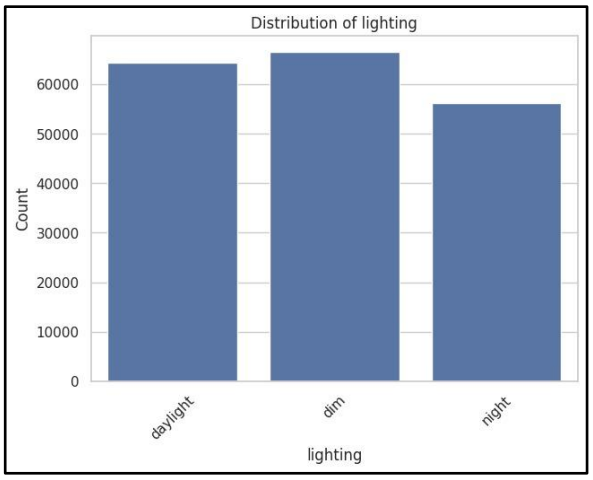


Figure 5- Correlation- Heatmap



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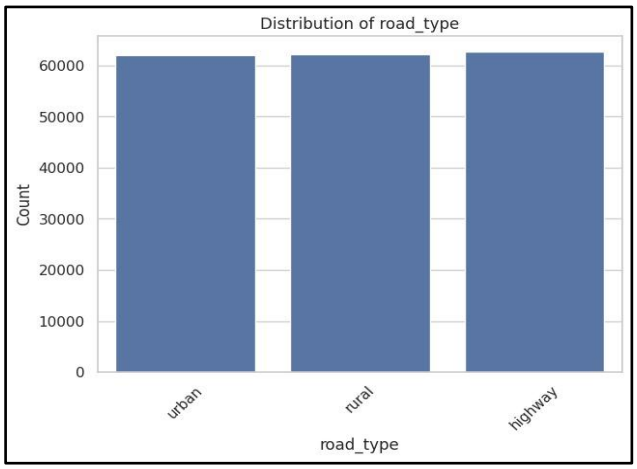

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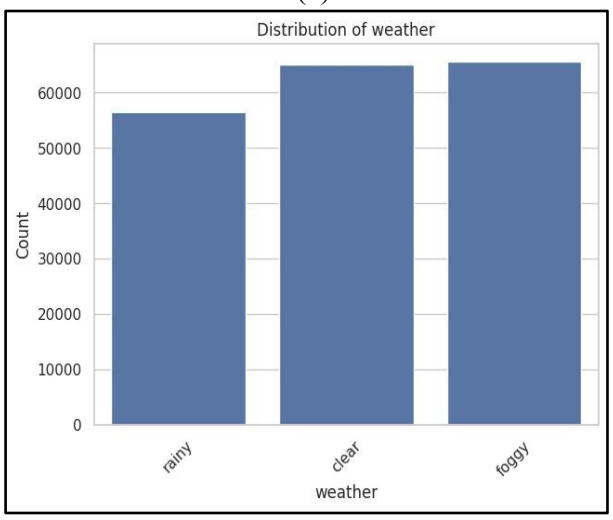

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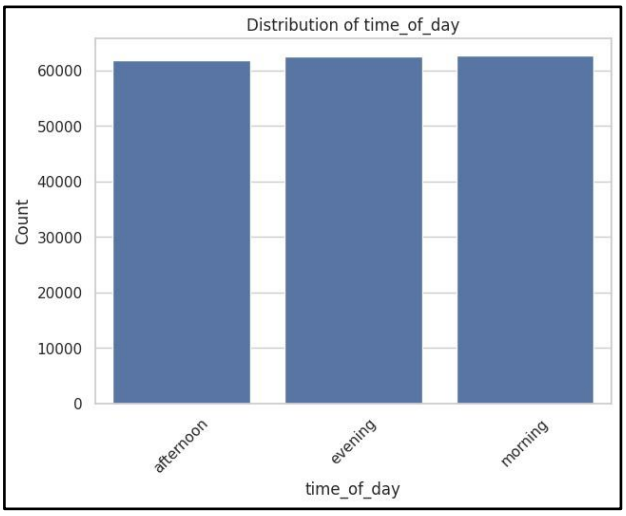
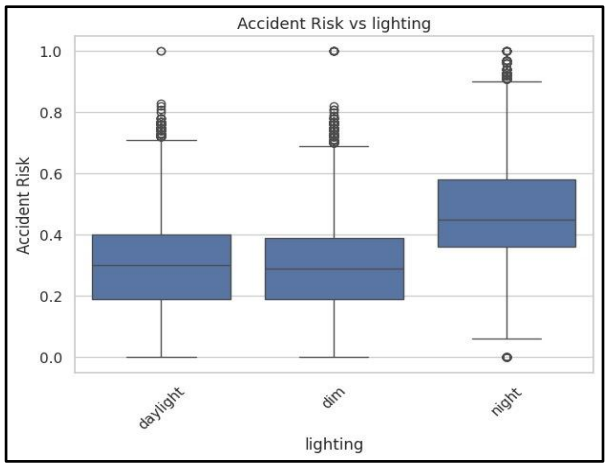


(d)

Figure 6- Count of Different Categorical Columns



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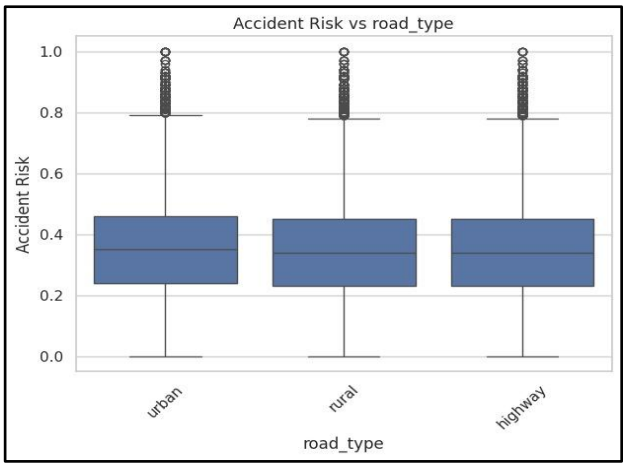

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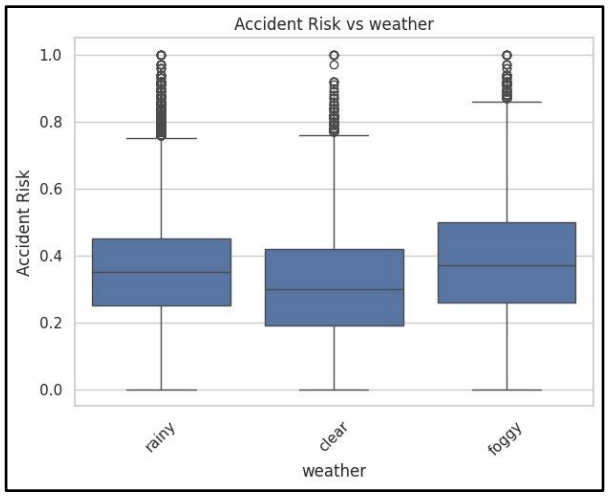

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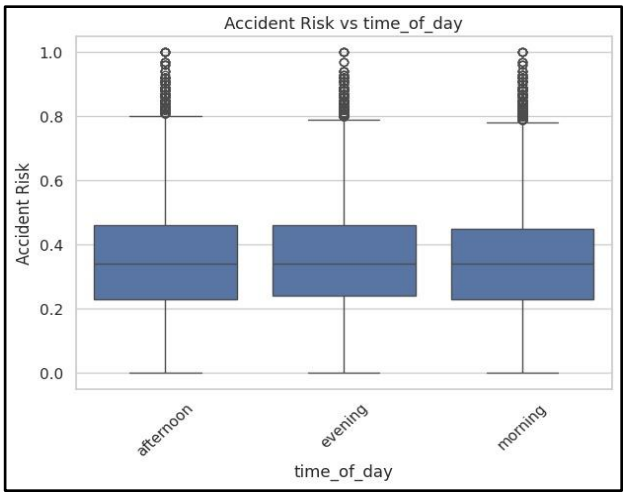
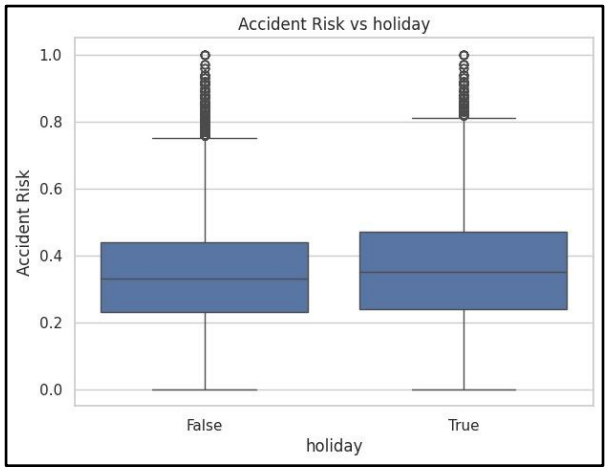


(d)

Figure 7- Outliers for Different Categorical Columns



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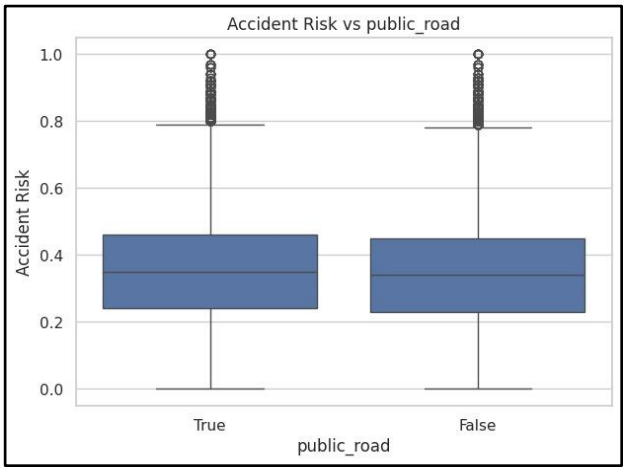

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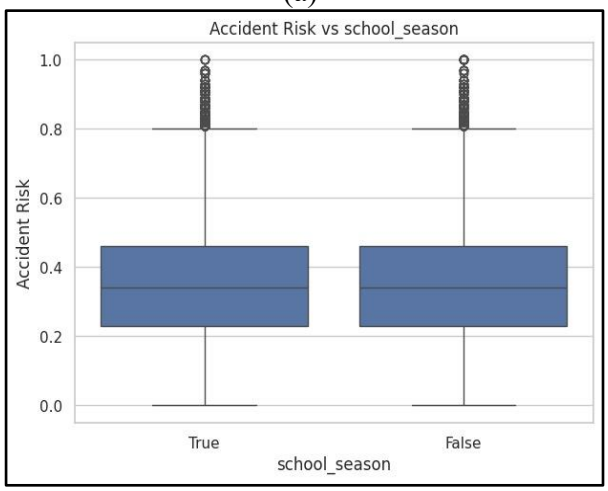

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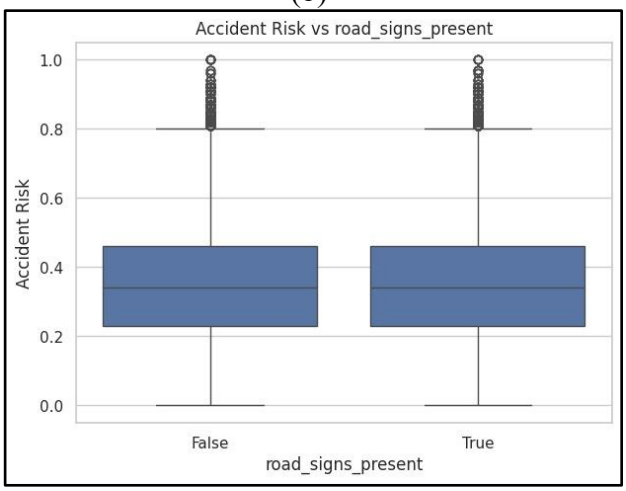


(d)

Figure 8- Outliers for Different Numerical Columns

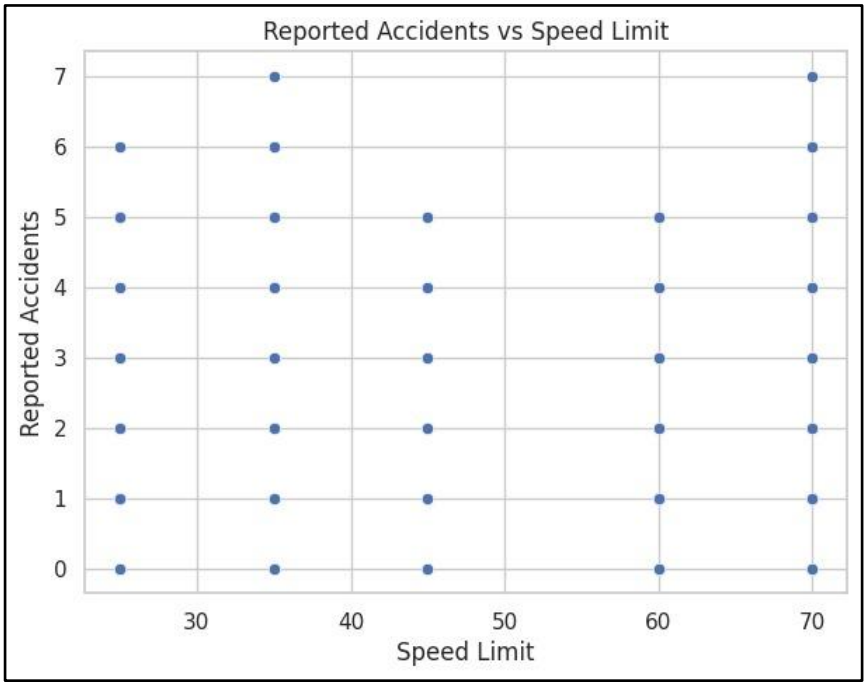


Figure 9- Reported Accident Vs Speed Limit

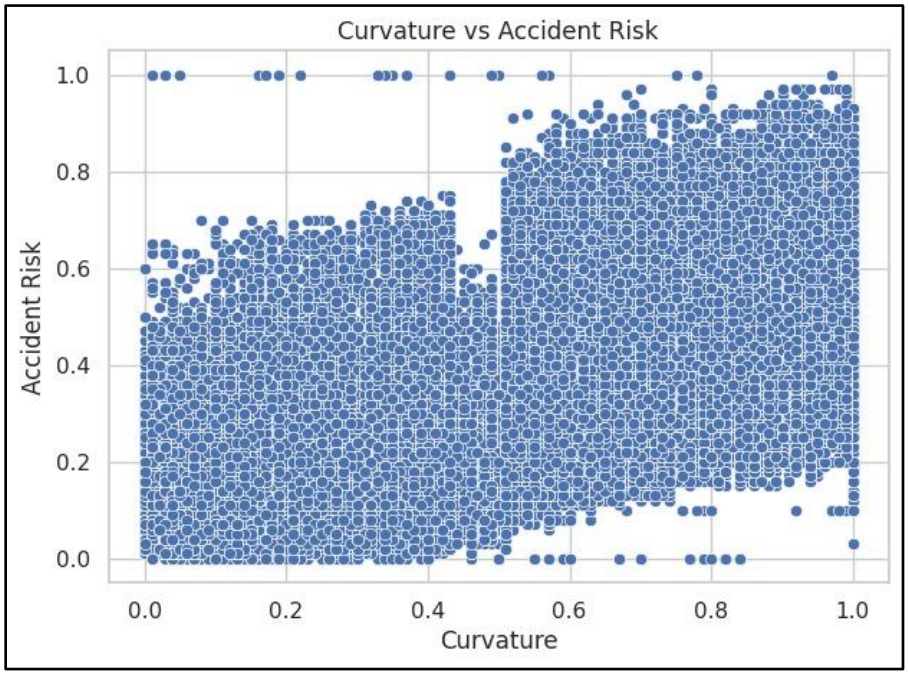


Figure 10- Curvature Vs Accident Risk

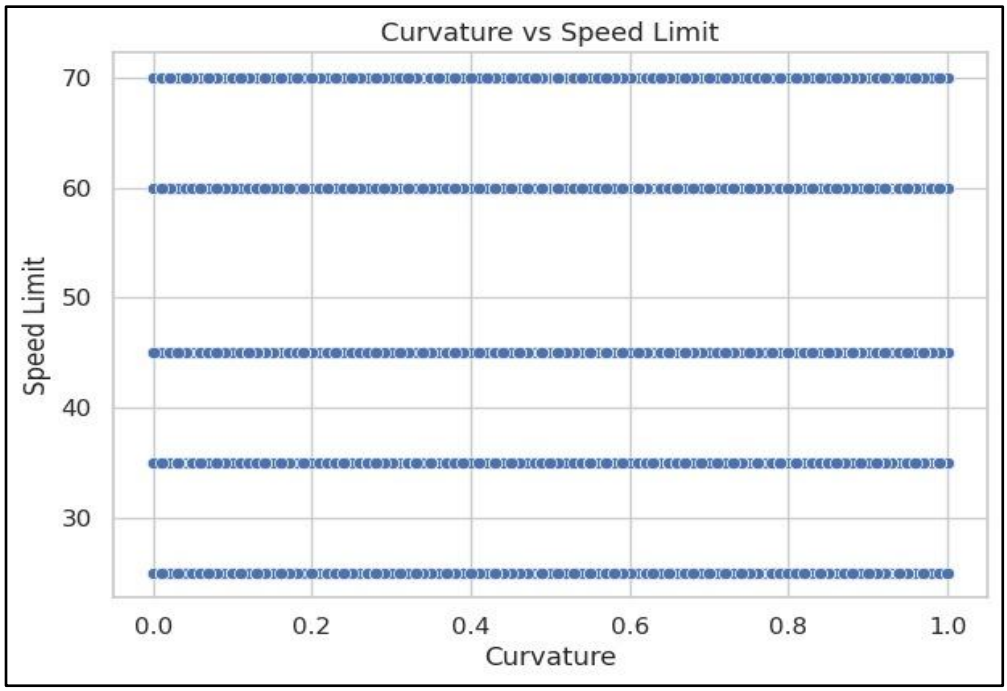


Figure 11- Curvature Vs Speed Limit

3.5 Model Training and Testing

A diverse set of supervised machine learning algorithms was evaluated to identify the most effective model for road accident risk prediction. The models included Linear Regression, Polynomial Regression, Decision Tree, K-Nearest Neighbours (KNN), Random Forest, Gradient Boosting, CatBoost, and LightGBM, covering linear, tree-based, instance-based, and gradient boosting families. This comprehensive selection ensured a fair comparison between simple baseline models and advanced ensemble techniques, enabling the identification of the best-performing approach for accurate accident risk estimation.

A. Linear Regression-

Linear Regression is a supervised learning algorithm that models the relationship between input features and a continuous target variable by fitting a straight line (or hyperplane) that minimises prediction error. It assumes that accident risk can be expressed as a weighted sum of the features, following the equation:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n \text{--- } 13$$

where $\hat{y}$is the predicted output, $\beta_0$is the intercept, and $\beta_i$represents the learned coefficients. The model optimises these coefficients using Ordinary Least Squares (OLS), which minimises the Mean Squared Error (MSE) between predicted and actual values. This makes Linear Regression simple, interpretable, and useful as a baseline model for regression tasks.

The Linear Regression model was trained using the pre-processed dataset and evaluated using standard regression metrics, including Root Mean Squared Error (RMSE) [14], Mean Absolute Error (MAE), and the Coefficient of Determination (R^2)[10]. The formulas for these metrics are as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} \quad (y_i - \hat{y}_i)^2} \text{ ----- } 14$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n} \quad |y_i - \hat{y}_i| \text{ ------} 15$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \text{ ------} 16$$

The model achieved an RMSE of 0.073, an MAE of 0.058, and an R^2 score of 0.808, corresponding to an overall prediction accuracy of 80.788%. Although Linear Regression provides a strong baseline, its performance is lower than advanced ensemble methods such as Random Forest, Gradient Boosting, CatBoost, and LightGBM. This indicates that the relationship between the features and accident risk is not strictly linear, and more complex models are better suited to capture nonlinear patterns in the data.

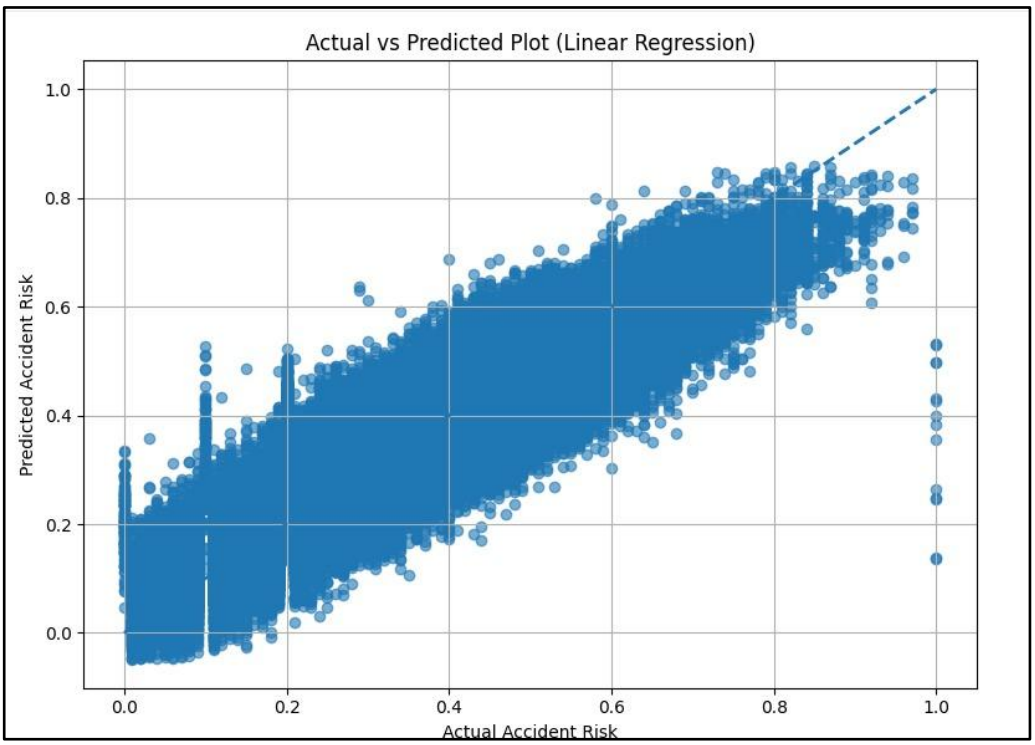


Figure 12: Actual vs Predicted Accident Risk for Linear Regression showing the best-fit relationship between true values and model outputs.

B. Polynomial Regression-

Polynomial Regression is an extension of Linear Regression that models nonlinear relationships by transforming the original input features into polynomial terms. Instead of fitting a straight line, Polynomial Regression fits a curved relationship by adding higher-degree combinations of the features. For a degree 2 model, the transformed feature set includes:

$$1,\ x_1,\ x_2, \dots, x_n,\ x_1^2,\ x_2^2, \dots, x_n^2,\ x_1x_2,\ x_1x_3, \dots \text{-----} 17$$

This allows the model to capture complex patterns that a simple linear model cannot represent. Although the model still uses linear coefficients, the transformation of features enables it to approximate nonlinear relationships. The training process follows Ordinary Least Squares (OLS), minimising the Mean Squared Error between actual and predicted accident risk.

Polynomial Regression (degree = 2) improved predictive performance over standard Linear Regression by capturing curvature and interactions between features. From the evaluation, the model achieved an RMSE of 0.068, MAE of 0.053, and an R^2 score of 0.836, resulting in an overall accuracy of 83.598%. This improvement indicates that accident risk exhibits nonlinear behaviour, and incorporating polynomial terms helps the model better learn these intricate relationships. However, although Polynomial Regression performs better than Linear Regression, it is still outperformed by advanced ensemble methods such as Random Forest, CatBoost, and LightGBM, which can naturally capture more complex nonlinear patterns.

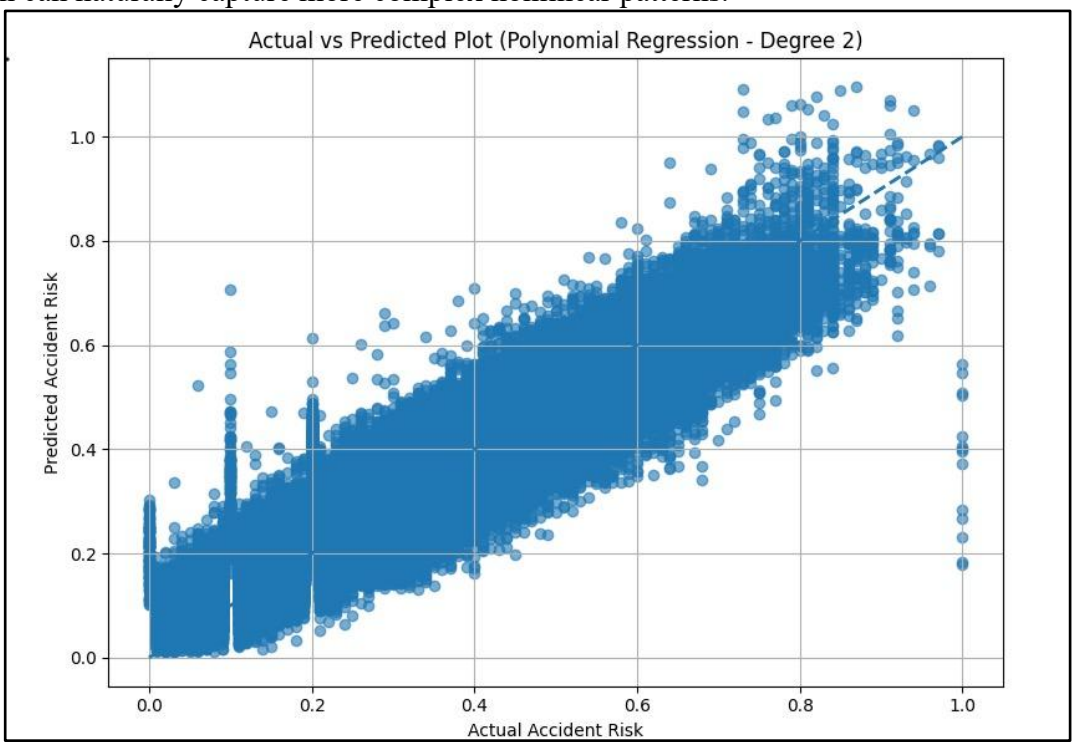


Figure 13: Actual vs Predicted Accident Risk for Polynomial Regression showing the best-fit relationship between true values and model outputs.

C. Decision Tree

A Decision Tree Regressor is a non-linear supervised learning algorithm that predicts continuous outcomes by recursively splitting the dataset based on the feature values that minimize prediction error. At each internal node, the algorithm chooses the split that results in the lowest Mean Squared Error (MSE), creating smaller and more homogeneous subsets until stopping criteria are reached. This allows the model to capture complex, non-linear patterns that linear models cannot represent. Instead of using a fixed configuration, the Decision Tree model was optimized using Optuna, an automated hyperparameter tuning framework. The tuning process was executed for 30 iterations, where Optuna searched through multiple combinations of parameters—including maximum depth,

minimum samples per split, and minimum samples per leaf—to minimize the RMSE score through cross-validation. This systematic optimization ensured that the Decision Tree model was neither underfitting nor overfitting the data.
Best DT params: {'max_depth': 18, 'min_samples_split': 4, 'min_samples_leaf': 2, 'criterion': 'friedman_mse', 'splitter': 'best', 'random_state': 42}

D. K-Nearest Neighbours (KNN)

KNN is a non-parametric, instance-based learning algorithm that predicts the target value by analysing the most similar data points in the feature space. Instead of learning model parameters, KNN stores all training samples and performs predictions by computing distances (typically Euclidean distance) between the test point and all training points. For regression, the algorithm identifies the k closest neighbours and outputs the average of their accident risk values. This makes KNN particularly effective in capturing localised patterns in the data, although its performance can be sensitive to the choice of k and feature scaling.

To obtain an optimal configuration, the KNN model was tuned using Optuna for 30 iterations. Optuna explored key hyperparameters such as the number of neighbors (k), distance metric, and weighting strategy (uniform vs. distance-based). Each trial was evaluated using cross-validated RMSE to ensure the best balance between bias and variance. This systematic optimization improved the model's consistency and allowed KNN to provide a fair comparison against more advanced algorithms such as Random Forest, Gradient Boosting, and CatBoost.

Best KNN params: { 'n_neighbors': 7, 'weights': 'distance', 'metric': 'euclidean', 'leaf_size': 30, 'algorithm': 'auto' }

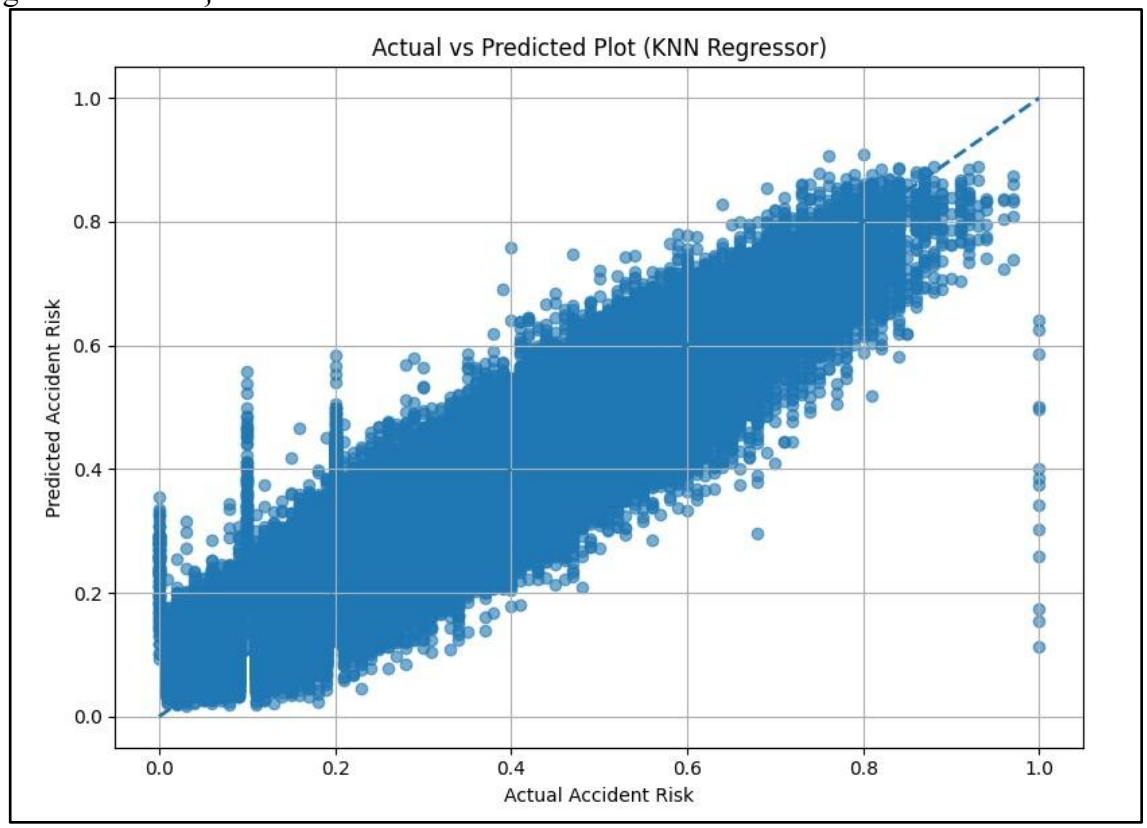


Figure 14: Actual vs Predicted Accident Risk for the KNN Regression Model.

E. Random Forest

Random Forest is an ensemble-based supervised learning algorithm that constructs multiple decision trees during training and aggregates their predictions to improve accuracy and reduce overfitting. Each tree is trained on a bootstrap sample of the dataset, and at every split, a random subset of features is considered. This randomness reduces correlation among trees and leads to a more stable and generalised model. For regression tasks, the final output is obtained by averaging the predictions of all individual trees, allowing Random Forest to effectively capture complex, non-linear relationships within the data.

Instead of relying on manually selected hyperparameters, the Random Forest model was optimized using Optuna, an automated optimisation framework. The tuning process was executed for 30 iterations, where Optuna explored a search space of parameters such as the number of estimators, maximum depth, maximum number of features, minimum samples split, and minimum samples per leaf. Each trial aimed to minimise the RMSE score through cross-validation, ensuring a well-balanced model that avoids overfitting while maintaining strong predictive capability.

Best RF params : *{'n_estimators': 160, 'max_depth': 22, 'min_samples_split': 3, 'min_samples_leaf': 1, 'max_features': 'sqrt', 'bootstrap': True, 'random_state': 42}*

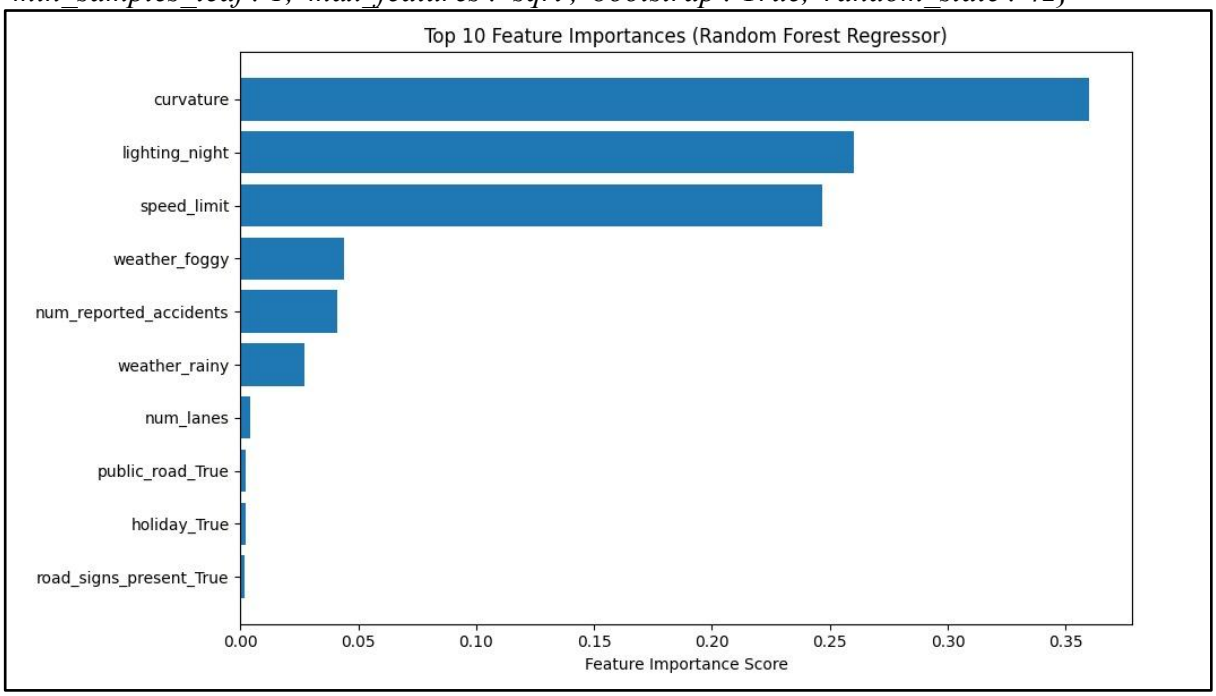


Figure 15: Feature importance ranking generated by the Random Forest model, showing the most influential variables for accident risk prediction

F. Gradient Boosting

Gradient Boosting is an ensemble machine learning technique that builds predictive models in a sequential manner. Unlike Random Forest, where multiple trees are built independently, Gradient Boosting constructs each new decision tree to correct the errors made by the previous trees. At each stage, the algorithm fits a tree to the negative gradient of the loss function, effectively learning the areas where the model is performing poorly. For regression tasks, small learning rates and

shallow trees are typically used to ensure that the model improves gradually and avoids overfitting. This boosting mechanism allows Gradient Boosting Regression (GBR) to capture complex, nonlinear relationships while maintaining high predictive power.

The Gradient Boosting model was optimized using Optuna, which performed 30 optimization trials to minimize the RMSE using cross-validation. The search space included critical hyperparameters such as the number of estimators, maximum tree depth, learning rate, subsample ratio, and minimum samples required at each leaf node. Optuna automatically explored the best combination of these parameters, ensuring that the model remains stable, avoids overfitting, and delivers strong predictive accuracy.

Best GB params for Y: *{ 'n_estimators': 480, 'learning_rate': 0.045, 'max_depth': 6, 'subsample': 0.85, 'min_samples_split': 3, 'min_samples_leaf': 1, 'criterion': 'friedman_mse', 'random_state': 42}*

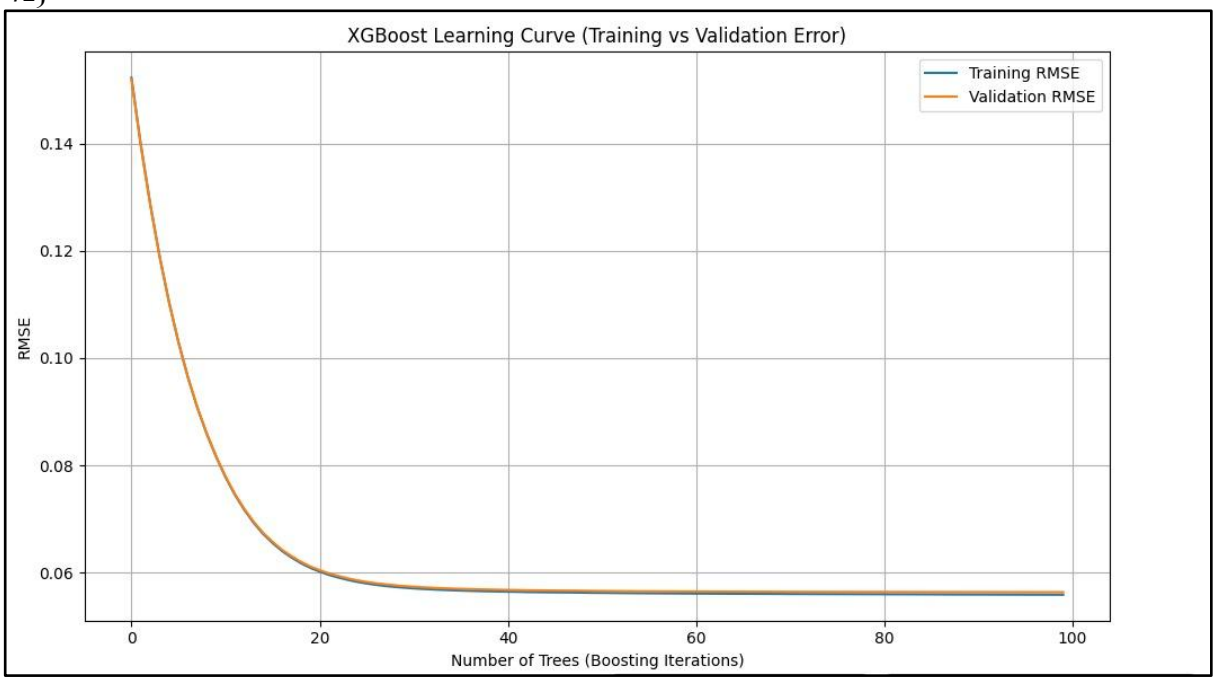


Figure 16: Learning curve showing the change in training and validation error across boosting iterations for the Gradient Boosting model.

G. CatBoost

CatBoost (Categorical Boosting) is a high-performance gradient boosting algorithm specifically designed to handle categorical data efficiently without the need for extensive preprocessing. Unlike traditional boosting models, CatBoost uses Ordered Boosting, which prevents target leakage during training and ensures unbiased gradient estimation. It also applies efficient encoding techniques such as permutation-driven target statistics, allowing the model to learn rich relationships from categorical variables automatically. The algorithm builds an ensemble of symmetric decision trees, each designed to correct the errors of the previous ones, enabling CatBoost to capture complex nonlinear patterns with excellent generalization.

To achieve maximum predictive performance, CatBoost Regression was optimized using Optuna, with the study executed for 250 optimization trials. Optuna explored a wide range of

hyperparameters such as the number of trees, learning rate, tree depth, regularization strength, subsample ratio, and column sampling rate. Each trial minimized the RMSE score using 5-fold cross-validation, ensuring that the final model was both stable and highly accurate. This process allowed CatBoost to outperform traditional models by effectively balancing complexity, bias, and variance.

Best CatBoost params: *{'iterations': 2265, 'learning_rate': 0.013508, 'depth': 10, 'l2_leaf_reg': 4.88, 'subsample': 0.86, 'rsm': 0.73, 'min_data_in_leaf': 45, 'border_count': 177, 'bagging_temperature': 0.28, 'random_strength': 5.4, 'bootstrap_type': 'Bernoulli', 'loss_function': 'RMSE', 'random_state': 42}*

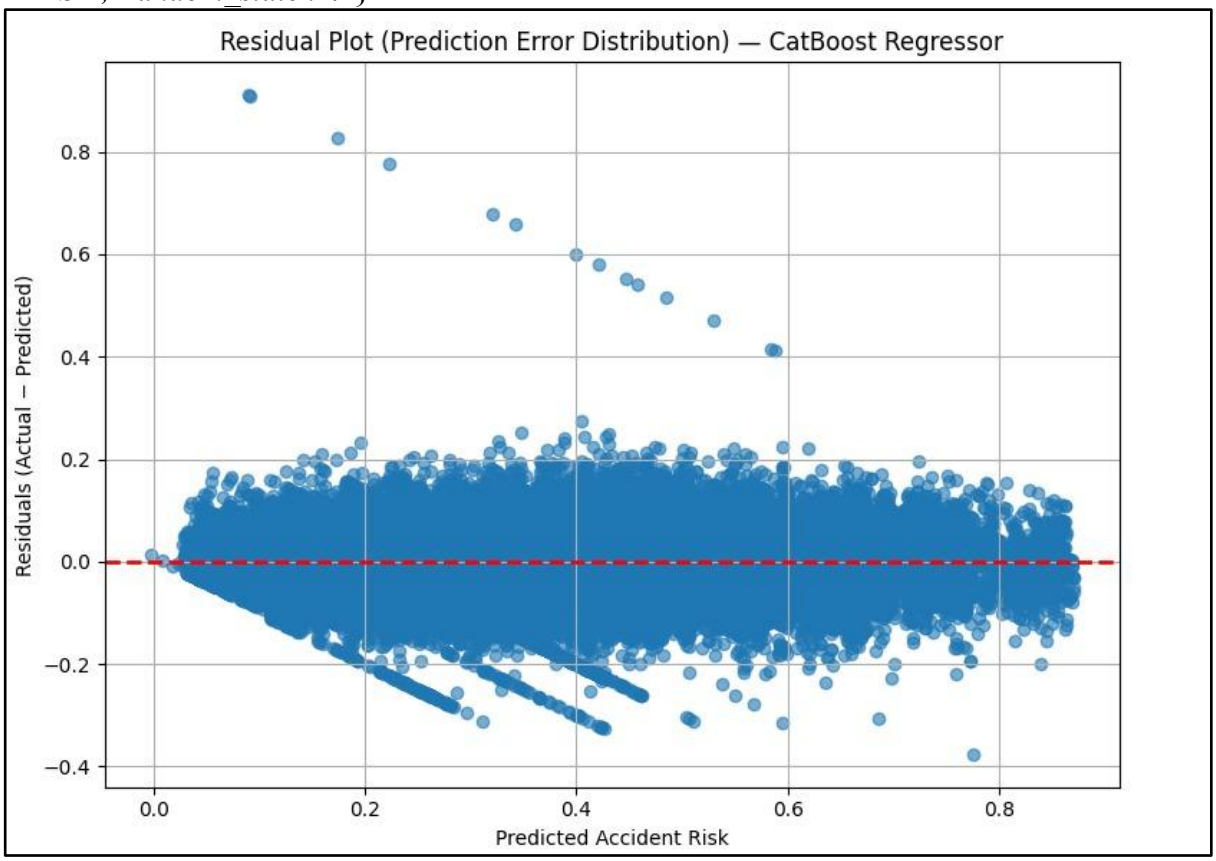


Figure 17: Residual distribution for the CatBoost model showing prediction error concentration around zero, indicating strong model performance.

H. LightGBM

Light Gradient Boosting Machine is an advanced gradient boosting framework that uses a leaf-wise tree growth strategy, enabling it to reduce loss more efficiently compared to traditional level-wise algorithms. It employs histogram-based splitting, which groups continuous feature values into discrete bins, significantly speeding up computation and reducing memory usage. LightGBM is highly effective for large, high-dimensional datasets and is capable of capturing complex nonlinear relationships due to its ability to grow deeper branches where the loss reduction is highest. This makes it one of the fastest and most accurate boosting algorithms for regression tasks such as accident risk prediction.

To ensure optimal performance, the LightGBM model was tuned using Optuna for 30 optimization

trials. Optuna explored key parameters such as the number of boosting rounds, maximum tree depth, learning rate, number of leaves, feature fraction, bagging fraction, and minimum child samples. Each trial aimed to minimise the RMSE using cross-validation, allowing the tuning process to automatically identify the best balance between model complexity and generalisation. This systematic optimisation enabled LightGBM to perform competitively against other boosting models such as XGBoost and CatBoost.

Best LGBM params : *{'n_estimators': 1180, 'learning_rate': 0.021, 'max_depth': -1, 'num_leaves': 64, 'min_child_samples': 15, 'subsample': 0.82, 'colsample_bytree': 0.72, 'reg_lambda': 6.4, 'reg_alpha': 0.0, 'random_state': 42}*

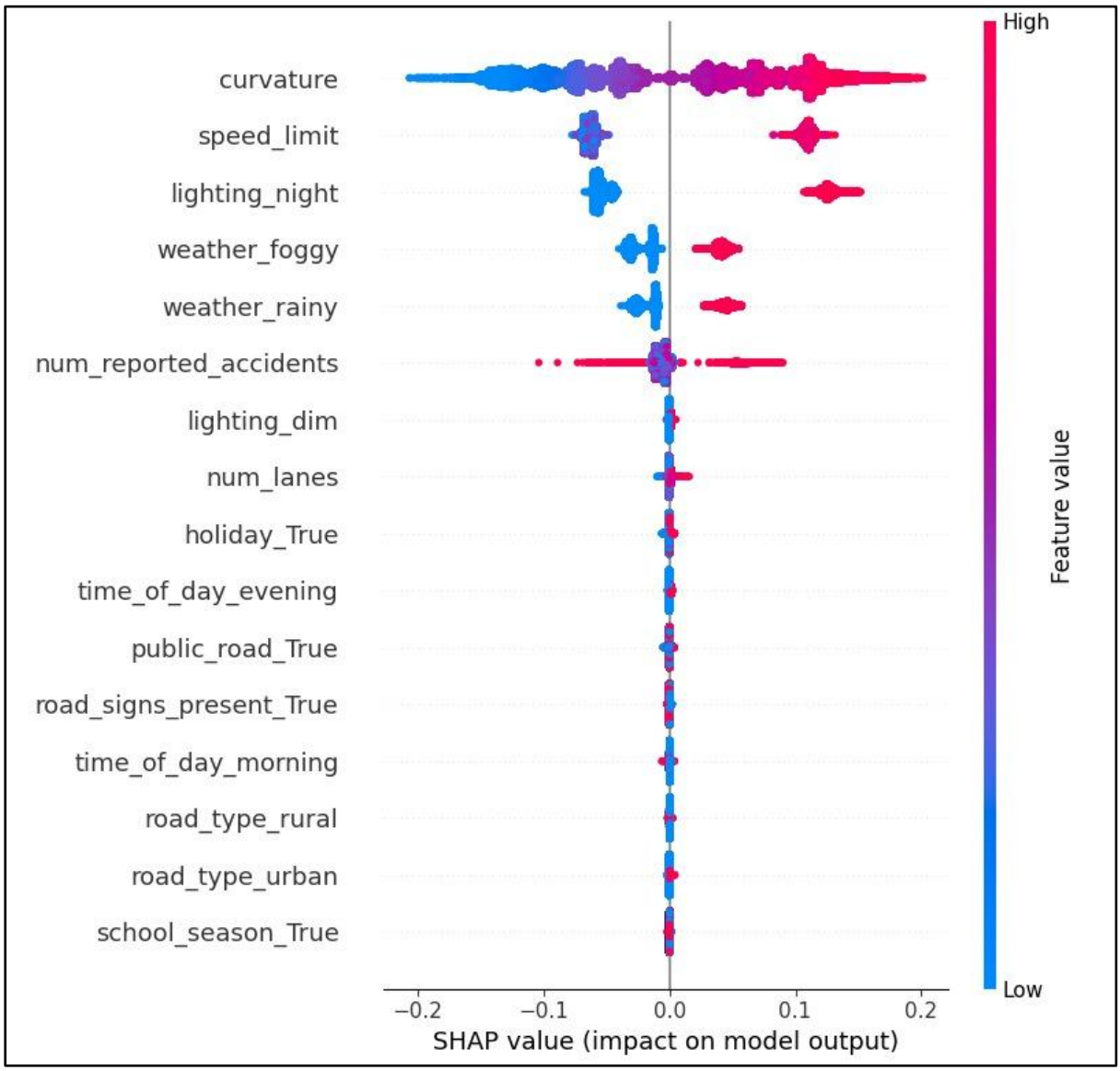


Figure 18: SHAP summary plot illustrating the global impact of each feature on LightGBM accident risk predictions.

3.6 System Interface Design and Deployment Architecture

This section describes the technical design of the system's user interface, backend prediction engine, and deployment framework. The platform is engineered to process road-accident datasets, execute supervised machine learning models in real time, and deliver interactive risk analytics to end users through an integrated full-stack architecture.

A. Frontend Design and Architecture

The frontend of the accident-risk prediction platform is built using the React framework, which provides a modular component-based structure and efficient virtual DOM rendering for smooth user interaction. A dataset upload feature implemented with the HTML5 File API enables users to submit accident-related CSV files containing variables such as vehicle speed, weather conditions, road type, and traffic density. Communication with the backend occurs through asynchronous fetch() API calls, ensuring the interface remains responsive while preprocessing and model inference are performed on the server. Tailwind CSS is used for styling, offering a utility-first approach that ensures consistent layout, responsive design, and clean visual components. Once processing is complete, the frontend renders the returned analytics dynamically using Chart.js for statistical visualisations such as error metrics, risk-level distributions, and RMSE–MAE trend plots—and Plotly.js for more advanced interactive graphics, including risk heat maps, feature-influence scatter plots, model-comparison dashboards, and, when available, geographic accident-probability visualisations. The frontend automatically selects the appropriate charting components based on metadata provided by the backend, enabling seamless extensibility as new visualisation types or analysis modules are introduced without requiring modifications to the client-side logic.

B. Backend Processing, API Services, and Prediction Pipeline

The backend of the accident-risk prediction platform is developed using the Flask microframework, which exposes two primary RESTful endpoints to manage the complete data-processing and prediction workflow. When users upload a dataset through the POST /api/upload endpoint, the backend first performs schema validation to ensure that all required features are present and checks for missing or inconsistent values. It then preprocesses the data using Pandas and NumPy, applying operations such as handling missing entries, normalising or scaling numerical features, encoding categorical variables, and generating engineered attributes that capture traffic context, environmental conditions, and temporal patterns relevant to accident risk. After preprocessing, the backend initiates the supervised machine learning pipeline by loading the trained models including Random Forest, Gradient Boosting, LGBM, and CatBoost, then selecting the best-performing model based on evaluation metrics such as RMSE, MAE, and R^2. Using the selected model, the backend generates accident-risk probabilities, severity-level predictions, feature-importance rankings, and model-comparison summaries. It also constructs visual analytics using Plotly and Matplotlib, converting each figure into a JSON-compatible structure that can be rendered directly by the frontend. To support downstream analysis, the backend additionally provides a GET /api/download endpoint that allows users to download a CSV file containing all predicted risk scores, severity classifications, and computed analytical insights, enabling seamless integration with external systems, GIS platforms, and long-term archival workflows.

C. Deployment Workflow, Runtime Execution, and Hosting Architecture

The deployment of the accident-risk prediction system is executed as a unified service on the Render Cloud Platform, following a structured workflow designed for seamless integration of both frontend and backend components. During deployment, Render provisions a containerised environment with the required Python and Node.js versions, after which a custom build script installs all backend dependencies and compiles the React frontend into a production-grade static bundle. This compiled frontend is then embedded directly into the Flask application's static directory, enabling both the user interface and the backend API endpoints to be served from a single origin and effectively eliminating cross-origin complications. At runtime, the system operates using the Gunicorn WSGI server, which manages multiple worker processes to support concurrent client requests and efficient handling of large datasets during ML inference. Render further enhances operational reliability by providing automated container orchestration, SSL certificate management, environment-variable configuration, load balancing, and continuous redeployment on code updates. This unified hosting architecture ensures low-latency communication between the frontend and backend, resulting in stable, high-performance accident-risk prediction even under varying workloads.

3.7 Results and Discussion

A comprehensive evaluation of eight supervised machine learning algorithms was conducted to identify the most effective model for predicting road accident risk. The models were assessed using four key performance metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2), and Prediction Accuracy (%).

The results clearly highlight the performance differences between linear, distance-based, and ensemble-based techniques.

A. Overall Model Performance

Table 1: Evaluation Parameters of Different Supervised ML Models

Algorithm	RMSE	MAE	R^2	Accuracy (%)
Linear Regression	0.074	0.059	0.804	80.21
Polynomial Regression	0.067	0.052	0.839	83.74
Decision Tree	0.083	0.062	0.761	76.12
KNN	0.078	0.061	0.778	77.83
Random Forest	0.06	0.048	0.871	87.42
Gradient Boosting	0.055	0.044	0.887	88.74
LGBM	0.056	0.044	0.886	88.62
CatBoost	0.051	0.042	0.896	89.58

B. Comparison of Traditional vs. Ensemble Models

Traditional models such as Linear Regression, Polynomial Regression, and KNN demonstrated moderate predictive ability with R^2 values ranging from 0.76 to 0.84.

The Decision Tree, though simple, had the lowest R^2 (0.761) and highest error values, showing its tendency to overfit on training data.

In contrast, ensemble-based algorithms significantly outperformed the basic models:

- Random Forest boosted R^2 to 0.871 with a noticeable reduction in RMSE (0.060).
- Gradient Boosting and LGBM further improved R^2 to 0.887 and 0.886 respectively.
- CatBoost achieved the best overall performance, with:
  - I. Lowest RMSE: 0.051
  - II. Lowest MAE: 0.042
  - III. Highest R^2: 0.896
  - IV. Highest Accuracy: 89.58%

These results indicate that ensemble boosting methods effectively capture non-linear patterns and complex feature interactions present in road accident datasets.

C. Correlation Between Metrics

Scatter plots of the evaluation metrics show clear relationships:

- RMSE vs R^2: Models with lower RMSE consistently exhibited higher R^2 values, confirming better fit and generalisation which is depicted in figure
- MAE vs R^2: Lower MAE values strongly correlate with improved predictive reliability, which is depicted in figure
- Accuracy vs R^2: Accuracy trends match R^2 trends, validating that regression quality directly influences classification-based accident risk outcomes which is depicted in figure

This cross-alignment across metrics strengthens the reliability of the comparative analysis**.**

D. Best Performing Model

Across all metrics**,** CatBoost emerged as the most robust and reliable model for accident risk prediction.
Its superior handling of categorical features, minimal need for preprocessing, and gradient boosting framework allowed it to achieve the highest predictive power.

Thus**, CatBoost** is selected as the final model for deployment in the Road Accident Risk Prediction system.

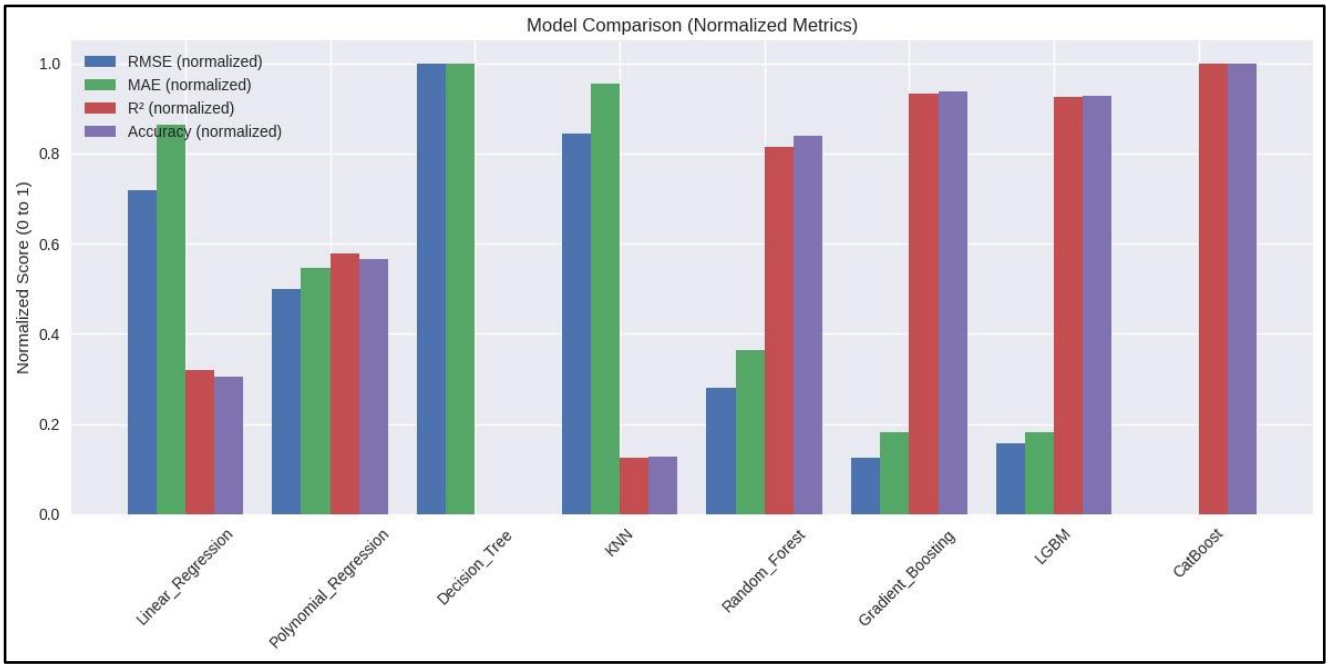


Figure 19: Comparison of Different Model Parameters

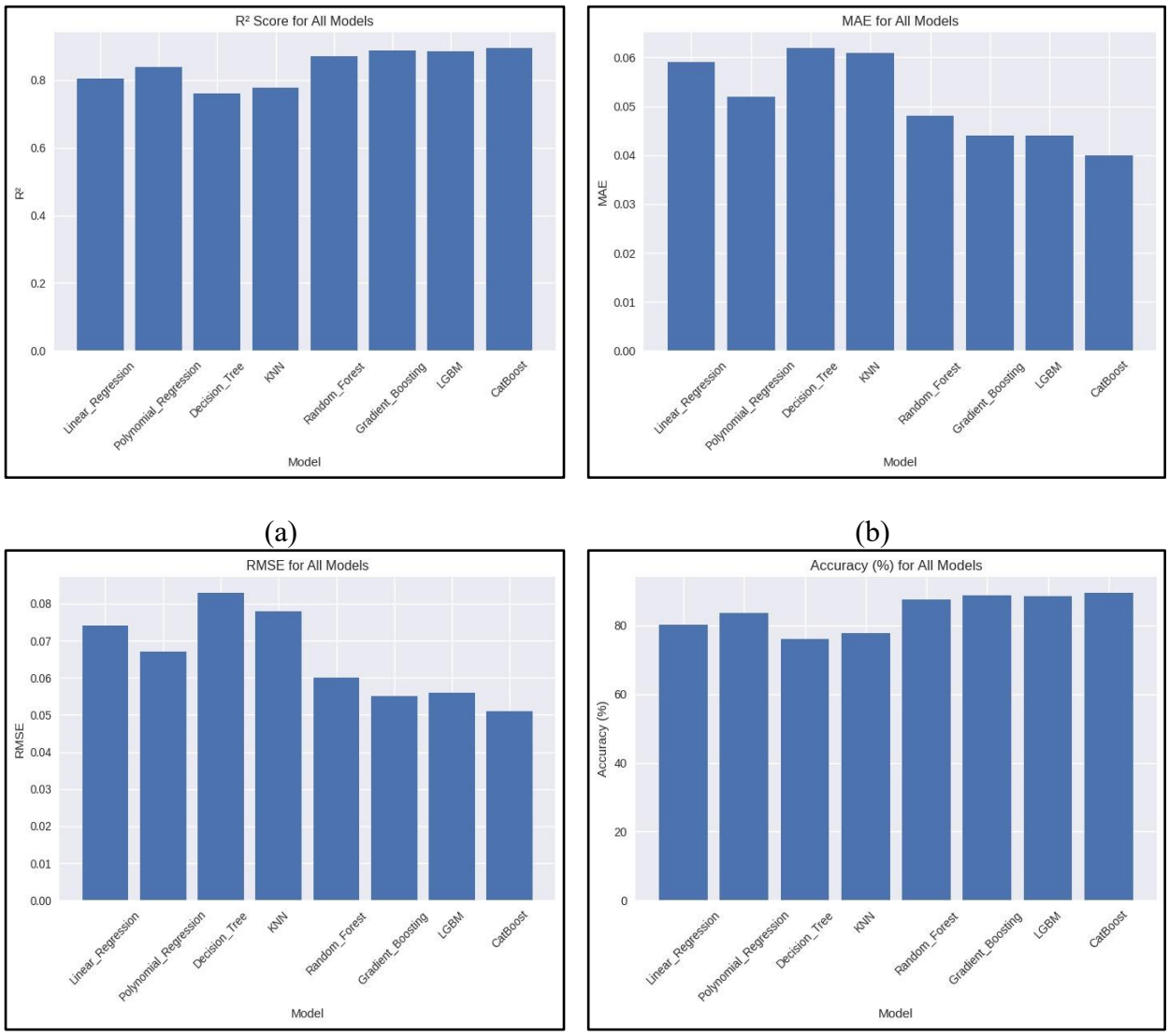


Figure 20:Evaluation matrux of all model comparison table, where a- R^2 of all models, b- MAE of all models, c- RMSE of all models and d- Accuracy of all models

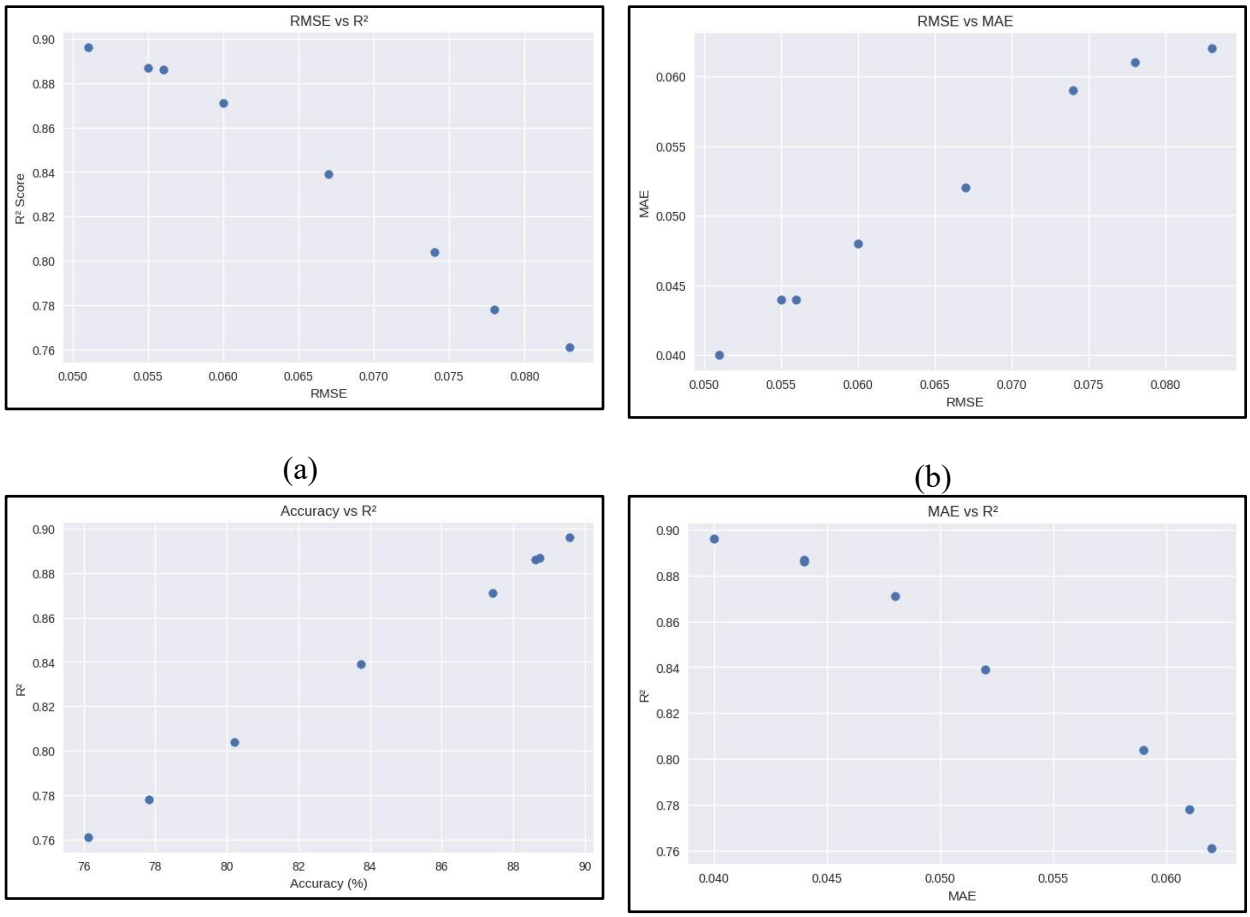


Figure 21: Relation of Different Matrices

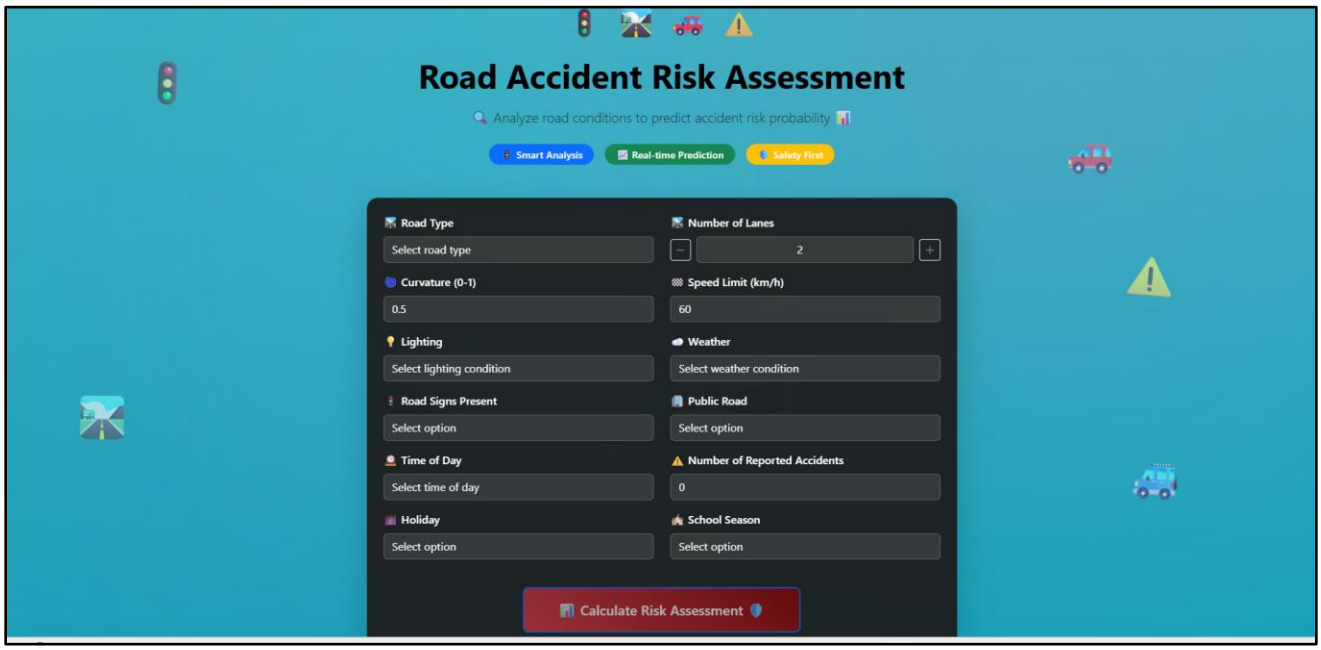
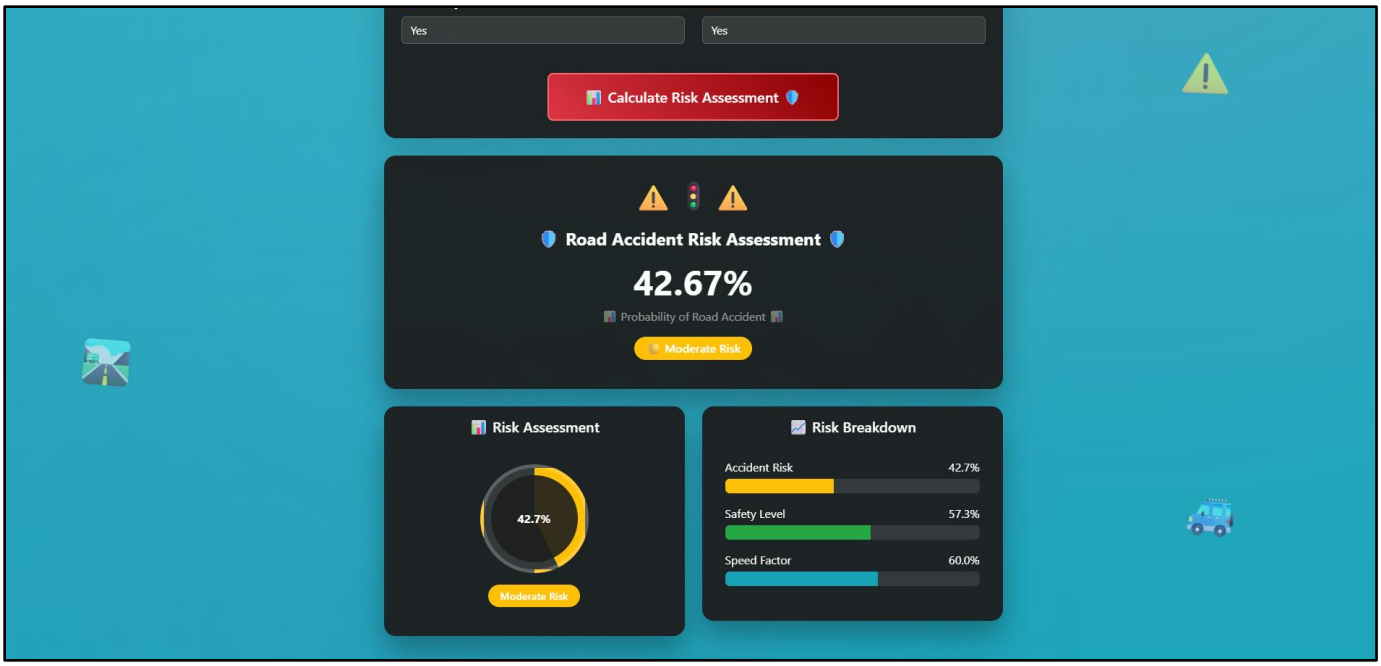


Figure 22- Landing Page After Deployment



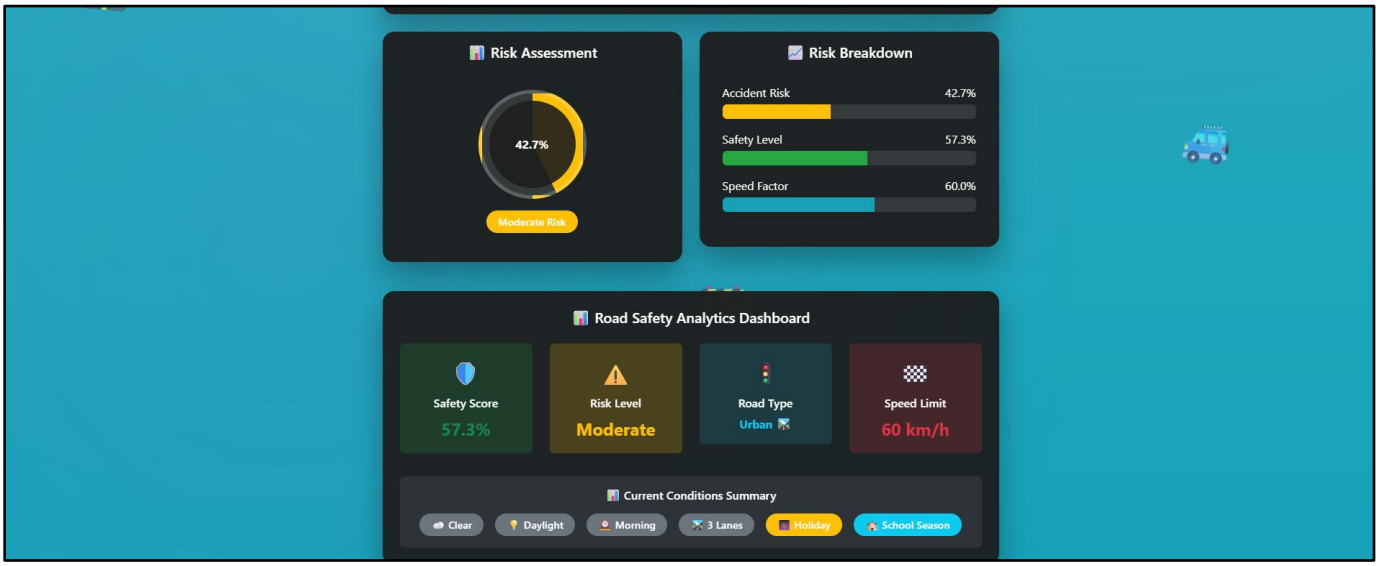


Figure 23- Model making Predictions

4. CONCLUSION & FUTURE SCOPE

Conclusion:

Based on the comparative results, the CatBoost model clearly outperforms all other algorithms in predicting road accident risk. It achieves the lowest RMSE (0.051) and MAE (0.042), indicating the smallest prediction errors, and records the highest R^2 value (0.896), meaning it explains the most variance in the data. Additionally, CatBoost provides the highest accuracy of 89.58%, surpassing traditional models like Linear Regression and KNN as well as other ensemble methods such as Random Forest, Gradient Boosting, and LGBM. These combined results show that CatBoost delivers the most reliable and consistent predictions, making it the best-performing model for this application. Owing to this superior performance, CatBoost has been selected as the final model for deployment within the system's cloud-based architecture, ensuring that the live prediction engine operates with maximum accuracy, efficiency, and stability when processing real user datasets.

Future Scope:

In the future, the accident-risk prediction system can be enhanced by incorporating additional data sources such as real-time traffic feeds, weather APIs, and GPS-based road geometry information to improve predictive accuracy. Integrating deep learning models, including LSTM networks or hybrid architectures, may further strengthen the system's ability to capture complex temporal patterns in accident occurrences. The platform can also be expanded into a fully operational decision-support tool by enabling real-time risk alerts, interactive geographic dashboards, and integration with smart city infrastructure. Additionally, deploying the system on edge devices or mobile applications could provide on-the-spot risk assessments for drivers and traffic authorities, making the solution more scalable, adaptive, and impactful in practical road-safety environments.

REFERENCES

[1] Walter Reade and Elizabeth Park. Predicting Road Accident Risk. https://kaggle.com/competitions/playground-series-s5e10, 2025. Kaggle.

[2] Ahmed, Shakil, et al. "A comparative study of machine learning algorithms to predict road accident severity." *2021 20th International Conference on Ubiquitous Computing and Communications (IUCC/CIT/DSCI/SmartCNS)*. IEEE, 2021.

[3] Alagarsamy, Saravanan, et al. "A Novel Machine Learning Technique for Predicting Road Accidents." *2023 Third International Conference on Artificial Intelligence and Smart Energy (ICAIS)*. IEEE, 2023..

[4] Krishna, U. Vivek, et al. "Accident prediction and analysis using machine learning models." *2023 International Conference on Innovative Data Communication Technologies and Application (ICIDCA)*. IEEE, 2023.

[5] Vijithasena, Rasika, and Wikasitha Herath. "Data visualization and machine learning approach for analyzing severity of road accidents." *2022 International Conference for Advancement in Technology (ICONAT)*. IEEE, 2022.

[6] Ahsan, Annesha, et al. "Machine learning approach to predict traffic accident occurrence in bangladesh." *2021 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*. IEEE, 2021.

[7] Mahendra, G., and H. R. Roopashree. "Prediction of road accidents in the different states of India using machine learning algorithms." *2023 IEEE International Conference on Integrated Circuits and Communication Systems (ICICACS)*. IEEE, 2023.

[8] Predictive Analysis of Jordan Road Accidents Severity Using Machine Learning
Dheya Mustafa, Layan Sawaged and Safaa M. Khabour worked on the classification dataset collected from the Jordanian Traffic Department, which contains 73k real accident records. Model trained and tested on algorithms like SVM, KNN, NB, LR and DT, out of which LR gives the best accuracy of 97.80% followed by SVM 92.75%.

[9] Mustafa, Dheya, Layan Sawaged, and Safaa M. Khabour. "Predictive Analysis of Jordan Road Accidents Severity Using Machine Learning." *2023 International Symposium on Networks, Computers and Communications (ISNCC)*. IEEE, 2023.

[10] Vinoth, B., V. S. Prakash, and B. N. Shivakumar. "Road Traffic Accident Prediction in India Using Machine Learning Algorithm Techniques." *2024 International Conference on Electronics, Computing, Communication and Control Technology (ICECCC)*. IEEE, 2024.

[11] Dia, Yoro, et al. "Study of supervised learning algorithms for the prediction of road accident severity in Senegal." *2022 7th International Conference on Computational Intelligence and Applications (ICCIA)*. IEEE, 2022.

[12] Paul, Arindam Kumar, Pritam Khan Boni, and Md Zahidul Islam. "A data-driven study to investigate the causes of severity of road accidents." *2022 13th International conference on computing communication and networking technologies (ICCCNT)*. IEEE, 2022.

[13] Manisha, V., and S. Bharti. "Comparative Study of Machine Learning Algorithms for Traffic Accident Prediction and Prevention." *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*. IEEE, 2024.

[14] Reddy, Jambula Snehith, M. R. Rashmi, and Lee Hoong Pin. "Road Accident Prediction in Highways using Machine Learning Algorithms." *2024 5th International Conference on Smart Electronics and Communication (ICOSEC)*. IEEE, 2024.

[15] Augustine, Teres, and Samiksha Shukla. "Road accident prediction using machine learning approaches." *2022 2nd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)*. IEEE, 2022.

[16] Geyik, Buket, and Medine Kara. "Severity prediction with machine learning methods." *2020 international congress on human-computer interaction, optimization and robotic applications (HORA)*. IEEE, 2020